



A2 Chemistry

12[a] Carboxylic Acids

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Candidates should be able to:

- ☐ Recall the reaction by which benzoic acid can be produced.
- ☐ Describe the reaction of carboxylic acids with PCl_3 , PCl_5 or $SOCl_2$ to form acyl chlorides.
- ☐ Understand and explain the further oxidation of some carboxylic acids.
- ☐ Describe and explain the relative acidities of carboxylic acids, phenols and alcohols.
- ☐ Describe and explain the relative acidities of chlorine-substituted carboxylic acids.

Definitions

Benzoic Acid	the simplest benzene-based carboxylic acid
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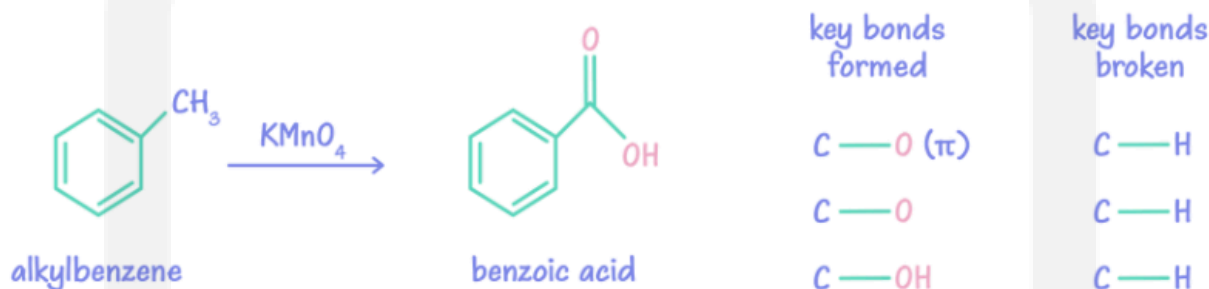
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12[a] Carboxylic Acids

12.1 Benzoic Acid

Benzoic Acid can be produced by the oxidation of the side chain of an alkyl benzene with hot alkaline potassium manganate, $KMnO_4$.

Conversion of alkylbenzenes to benzoic acids using $KMnO_4$



12.2 Further Oxidation of Carboxylic Acids

Methanoic acid does not have an alkyl chain attached to the $-COOH$ group, instead a hydrogen is attached. Therefore, it has some reactions in common with aldehydes, in particular, methanoic acid is oxidised to Carbon Dioxide and water, with common oxidising agents, such as $K_2Cr_2O_7$, $KMnO_4$, Fehling's reagent or Tollens' reagent.

Ethanedioic acid contains two adjacent carbonyl groups. The proximity of two δ^+ carbon atoms weakens the $C-C$ bond sufficiently for the molecule to be readily oxidised by warm acidified manganate(VII).



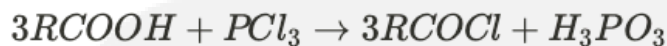
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12.3 Reactions of Carboxylic Acid

Carboxylic acid forms acyl chlorides by reacting with the following reagents:

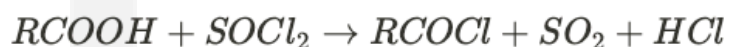
- With phosphorus trichloride:-



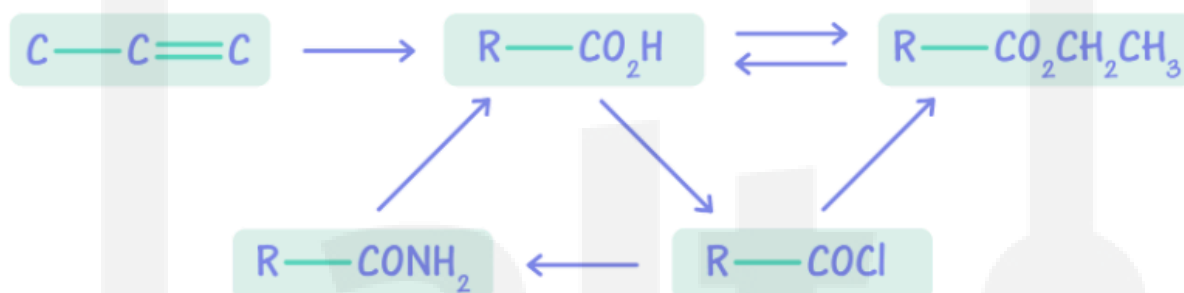
- With phosphorus pentachloride:-



- With thionyl chloride:-



Further reactions of carboxylic acids are summarized below:



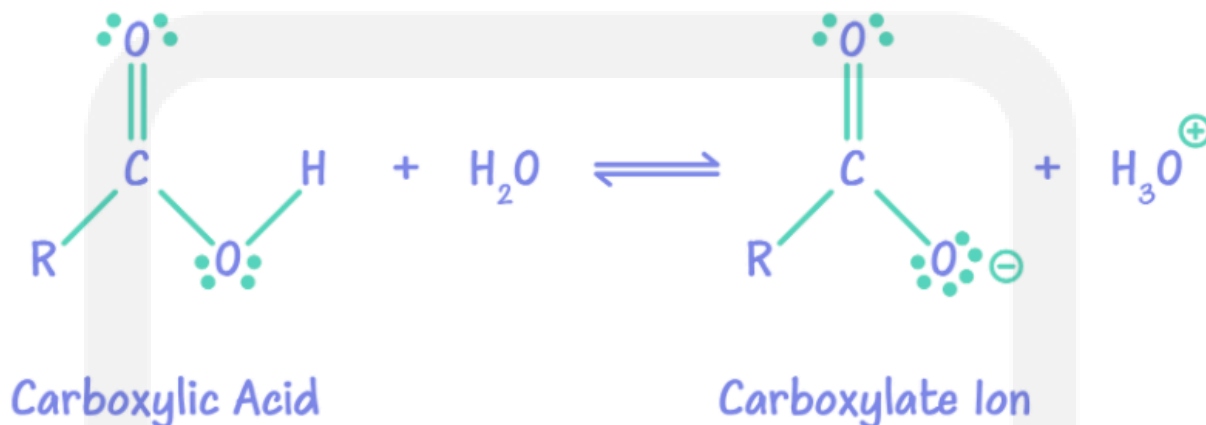


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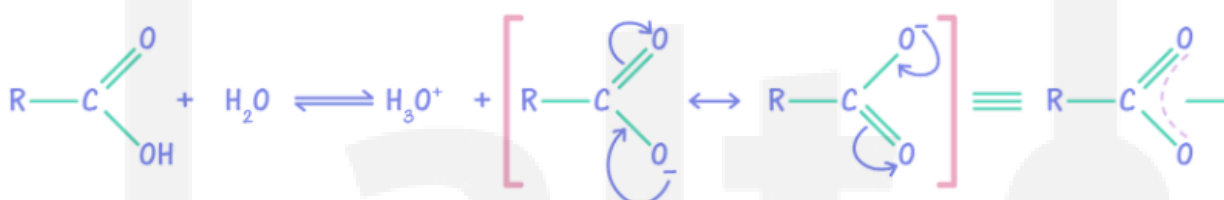
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12.4 Acidity

Carboxylic acids are weak acids that dissociate partially in aqueous solution according to the equation.



Carboxylic acids are more acidic than alcohols because the negative charge on the anion can be delocalised over two electronegative oxygen atoms.



Electrons in the C-O bond are drawn towards the C=O bond, and electrons are drawn away from the O-H bond. And the negative charge is spread over the whole -COO^- group.

12.4.1 Relative Acidities

Carboxylic acids are more acidic than phenols and alcohols. The relative acidity is Carboxylic acids > Phenols > Alcohols. This can be explained by the increasing ability of the molecular structures to delocalise the negative charge in the alkoxide, phenoxide and carboxylate anions. Atoms or groups that draw electrons away from the -COO^- group will help the anion to form, and this causes the acid to be more dissociated. On the other hand, groups that donate electrons to the -COO^- group will cause the acid to become weaker.

Electron donating groups will decrease the acid strength of carboxylic acids, whereas electron withdrawing groups will increase the acid strength of carboxylic groups.

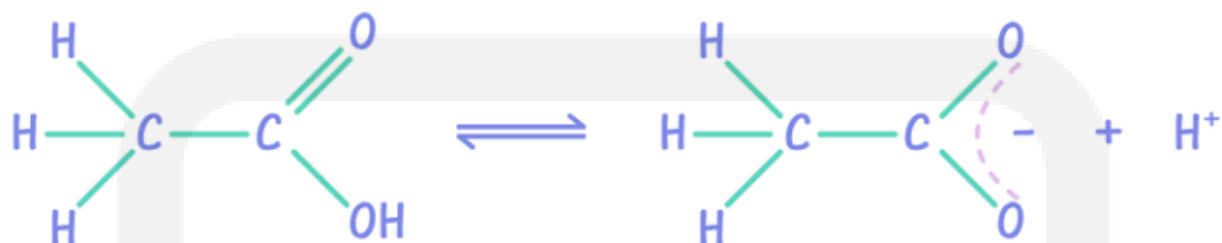


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12.4.2 Chloroethanoic Acid vs Ethanoic Acid

Chloroethanoic is a stronger acid than ethanoic acid.



Chloroacetic Acid

Conjugate Base

The difference in acidity can be explained by considering the stability of the conjugate base (anion) formed. The more stable the anion is, the more likely is its formation.

The chlorine atom is very electronegative, and in chloroethanoate ion, it pulls the electron density away from the -COO^- group, making it less attracted to the H^+ ion. The conjugate base is, therefore, more stable. A more stable conjugate base indicates a greater tendency to dissociate and, therefore, a stronger acid.



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12.4.3 Trichloroethanoic Acid vs Chloroethanoic Acid

Trichloroethanoic acid is stronger than chloroethanoic acid.

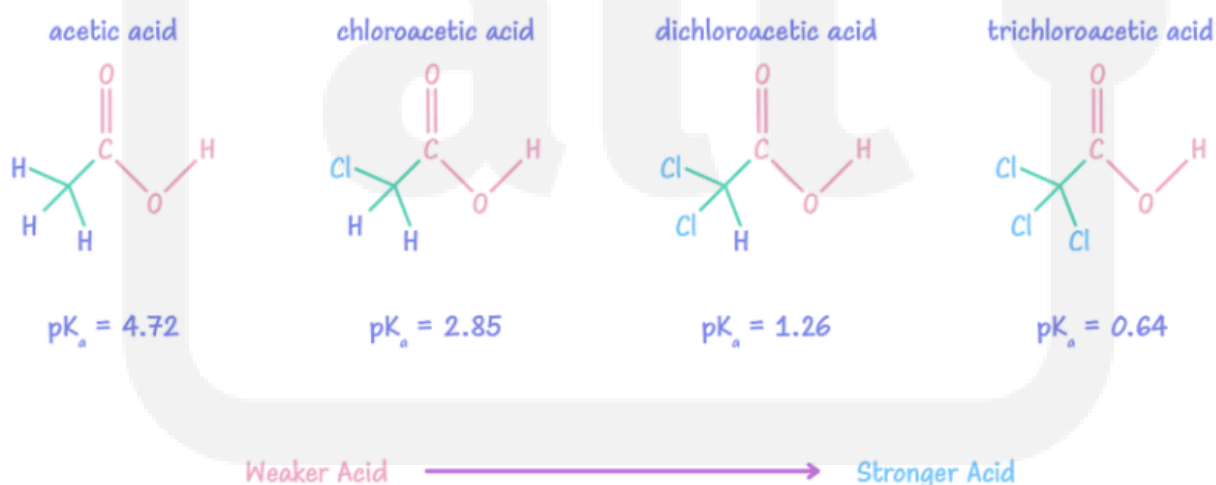


Strong Acid $pK_a = 0.63$

Weak Conjugate Base (Stabilized)

Three electronegative Cl atoms have a greater electron withdrawing effect than just one. More electron density is pulled away from the COO^- group making it less negative and less likely to attract back the H^+ ions. Thus, trichloroethanoic acid is more stable than chloroethanoic acid, and it dissociates more.

Substituent Effect on Acidity





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12.4.4 Effect of distance of Cl atom on acidity

2-chlorobutanoic acid is a stronger acid than 4-chlorobutanoic acid. The further the Cl is from the COO^- group in the anion, the lesser the electron withdrawing effect it has on it.

Points to Note:

- Methanoic acid can be oxidised by Fehling's reagent and Tollens' reagent, but ethanoic acid cannot be oxidised by these reagents.
- Relative acidity is Carboxylic acids > Phenols > Alcohols.
- As the distance of chlorine atoms from the COO^- group increases, the acidity decreases as the electron withdrawing effect decreases.

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